

2016.0 RANGE ROVER (LG), 303-04

FUEL CHARGING AND CONTROLS - TURBOCHARGER - TDV6 3.0L DIESEL

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the fuel charging and controls system and operation, refer to the relevant Description and Operation section of the workshop manual. REFER to: [Turbocharger](#) (303-04B Fuel Charging and Controls - Turbocharger - TDV6 3.0L Diesel, Description and Operation).

INSPECTION AND VERIFICATION

WARNING:

The following tests may involve working in close proximity to hot components. Make sure adequate protection is used. Failure to follow this instruction may result in personal injury.

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle.

NOTE:

Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Intake air system ▪ Hose(s) / hose connections ▪ Vacuum hoses ▪ Turbine intake shut-off valve ▪ Charge air shut-off valve ▪ Charge air radiator ▪ Turbocharger - primary/secondary ▪ General engine condition	▪ Fuses ▪ Electrical connections and harnesses ▪ Turbocharger actuators ▪ Turbine intake shut-off valve ▪ Turbine intake shut-off valve solenoid ▪ Charge air shut-off valve solenoid ▪ Engine Control Module (ECM)

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. If the cause is not visually evident, verify the symptom and refer to the

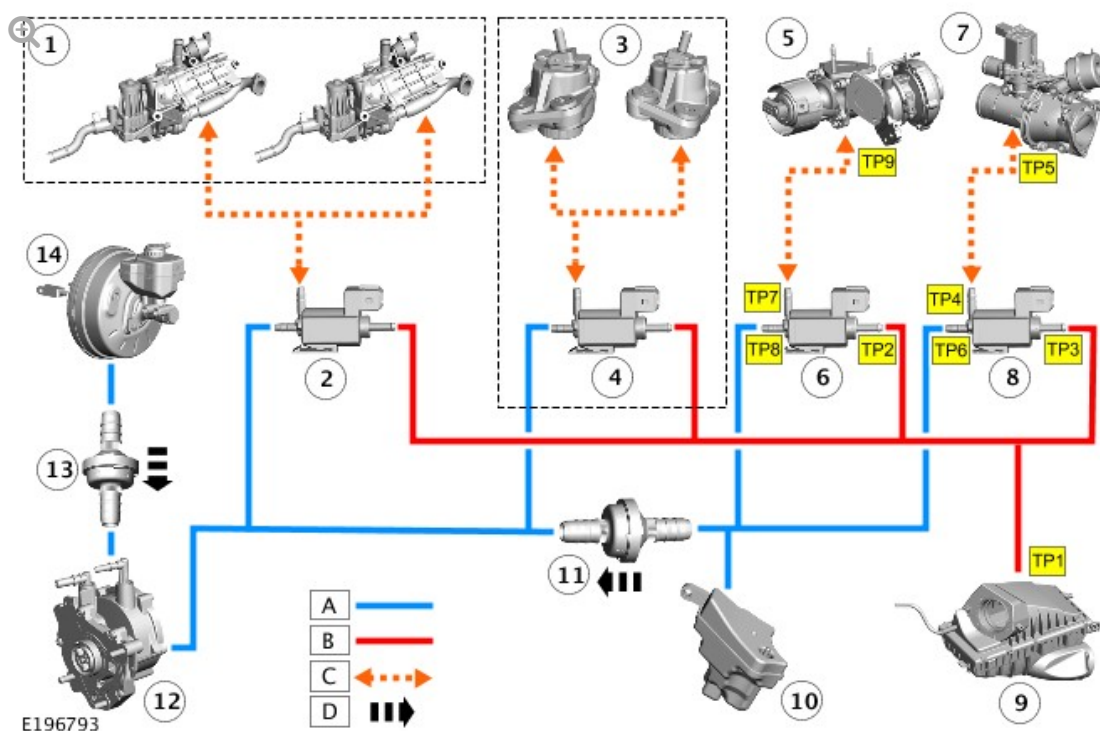
Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

5. Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Poor performance	▪ Low/Contaminated fuel ▪ Restricted intake air system ▪ General engine condition ▪ Engine control module failure	Check the fuel level and condition. Draw off approximately 1 ltr (2.11 pints) of fuel and allow to stand for 1 minute. Check to make sure there is no separation of the fuel indicating water or other liquid in the fuel. Check the intake air system for restriction. Check the engine condition, compressions, etc if there are indications of a mechanical fault. Check for DTCs indicating a module fault. Refer to the warranty policy and procedures manual if an engine control module is suspect.
No boost	▪ Electrical connections and harnesses ▪ Restricted intake air system ▪ Charge air cooler restricted/leaking ▪ Turbocharger actuator failure(s) ▪ Turbocharger failure(s) ▪ Engine control module failure	Check the electrical connections and harnesses. Check the intake air system for restriction/leakage. Check the turbocharger actuators and circuits. Refer to the electrical guides. Check the turbochargers for wear. Disconnect the turbocharger intake and outlet pipework and turn the turbochargers by hand. Any roughness indicates a fault. Check any up and down movement in the turbocharger shafts. Excessive movement indicates a fault. If in doubt, compare the suspect unit with a new turbocharger. Check for DTCs indicating an actuator or module fault. Refer to the warranty policy and procedures manual if an engine control module is suspect.
No boost/excessive noise	▪ Turbocharger failure(s)	Disconnect the turbocharger intake and outlet pipework and turn the turbochargers by hand. Any roughness indicates a fault. Check any up and down movement in the turbocharger shafts. Excessive movement indicates a fault. If in doubt, compare the suspect unit with a new turbocharger.

TDV6 3.0L DIESEL ENGINE - VACUUM SYSTEM LAYOUT. TWIN TURBO VARIANTS ONLY



ITEM	DESCRIPTION
1	High pressure EGR(s) (1 or 2 may be fitted)
2	EGR Cooler bypass valve solenoid
3	Active engine mounts (x2) (where fitted)
4	Active engine mount solenoid (where fitted)
5	Turbine intake shut-off valve
6	Turbine intake shut-off valve solenoid
7	Charge air shut-off valve
8	Charge air shut-off valve solenoid
9	Air filter housing
10	Vacuum reservoir
11	Non return valve
12	Vacuum pump

13	Non return valve
14	Brake servo
A	Vacuum supply hose
B	Vacuum vent hose
C	Actuator vacuum supply and vent hose
D	Direction of flow
TP	Pin point test reference - Test Point

PINPOINT TESTS

NOTE:

Refit any hoses that have been disconnected before moving on to the next step

PINPOINT TEST A : VACUUM CONTROL SYSTEM TESTS - TWIN TURBO VARIANTS ONLY

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: VACUUM VENT HOSE TEST 1	
	1 With the ignition switched off, disconnect the vacuum vent hose from the air filter housing at point TP1
	2 Connect a hand operated vacuum pump into the circuit at point TP1 and apply a vacuum
	Can a vacuum be achieved and maintained? Yes GO to A2. No Suspect air leak, carry out a smoke test to identify the leak point. If a leak is identified, repair fault and repeat step 1 to confirm repair
A2: VACUUM VENT HOSE TEST 2	
	1 Disconnect the vacuum vent hose from the turbine intake shut-off valve solenoid at point TP2
	2 Connect the vacuum pump to the vent hose at point TP2 and apply a

	vacuum
	Can a vacuum be achieved? Yes Check the vacuum vent hose for a blockage between the turbine intake shut-off valve solenoid and the air filter housing No If a vacuum cannot be achieved the vent hose is functioning correctly Repeat the test for the charge air shut-off valve vent hose connecting the gauge at point TP3 If the turbine intake shut-off valve and the charge air shut-off valve vent hoses are functioning correctly GO to A3.

A3: TURBINE INTAKE SHUT-OFF VALVE CLOSED POSITION CHECK

	1 Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signals - Turbo Shut-off valve opening position - Measured (0x03F1) and Turbo Shut-off valve opening position - Commanded (0x03F0)
	With the ignition on and the engine off is the value for signal Turbo Shut-off valve opening position - Measured <4%? Yes GO to A4. No Suspect turbine intake shut-off valve is faulty and sticking open Install a new turbine intake shut-off valve After installing a new turbine intake shut-off valve, carry out the calibration procedure. Using the Jaguar Land Rover Approved Diagnostic Equipment, perform routine - Powertrain set-up – Turbine shut-off valve

A4: CHARGE AIR SHUT-OFF VALVE OPERATION CHECK 1

	1 Connect a hand operated vacuum pump to the charge air shut-off valve hose at point TP4
	2 Check the operation of the charge air shut-off valve by gradually applying a vacuum and then releasing it while observing the charge air shut-off valve actuator connecting rod
	3 Note: A clunk can be heard from the charge air shut-off valve as it closes
	Did the charge air shut-off valve open as a vacuum was applied and close when the vacuum was released? Yes GO to A6. No GO to A5.

A5: CHARGE AIR SHUT-OFF VALVE OPERATION CHECK 2

	1 Connect a hand operated vacuum pump directly to the charge air shut-off valve at point TP5

	2 Check the operation of the charge air shut-off valve by gradually applying a vacuum and then releasing it while observing the charge air shut-off valve actuator connecting rod
	3 Note: A clunk can be heard from the charge air shut-off valve as it closes
	Did the charge air shut-off valve open as a vacuum was applied and close when the vacuum was released? Yes Suspect blockage in the vacuum hose between the charge air shut-off valve solenoid and charge air shut-off valve Inspect vacuum hose for blockages/defects and rectify as required No Suspect a faulty charge air shut-off valve Install a new charge air shut-off valve

A6: CHARGE AIR SHUT-OFF VALVE ACTIVATION CHECK 1

	1 Connect a hand operated vacuum pump using a 'T' piece between the charge air shut-off valve and the charge air shut-off valve solenoid at point TP4
	2 Start the engine and allow to idle. View the vacuum reading while in mono turbo mode (during the first three minutes of idle)
	Is a vacuum present? Yes Remove and inspect the charge air shut-off valve solenoid. Suspect solenoid stuck open. Replace as required No GO to A7.

A7: CHARGE AIR SHUT-OFF VALVE ACTIVATION CHECK 2

	1 With the hand operated vacuum pump still connected at point TP4 via the "T" piece
	2 Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signals - Turbo Shut-off valve opening position - Commanded (0x03F0) and Turbo Shut-off valve opening position - Measured (0x03F1)
	3 After approximately 3 minutes at idle the turbine intake shut-off valve and charge air shut-off valve will automatically be triggered to open and activate the secondary turbo
	4 This can be confirmed by datalogger signal - Turbo Shut-off valve opening position - Commanded changing value to 100%
	5 When this occurs a vacuum of approximately 800mbar should be displayed on the gauge
	6 Observe the value on the vacuum gauge as the secondary turbo is activated

	Is the correct vacuum achieved and maintained? Yes GO to A8. No GO to A9.
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A8: CHARGE AIR SHUT-OFF VALVE ACTIVATION CHECK 3

	1 With the engine idling in bi-turbo mode an increase in engine speed will return the engine to mono turbo mode
	2 Increase engine speed and then allow the engine to return to idle
	3 The vacuum will return to zero (within 2-3 seconds)
	4 Observe the value on the vacuum gauge as the secondary turbo is deactivated
	Is the correct vacuum is achieved and maintained? Yes GO to A10. No Check that the datalogger signal - Turbo Shut-off valve opening position - Commanded (0x03F0) has changed to 0%. If so, remove and inspect the charge air shut-off valve solenoid. Suspect solenoid stuck open. Replace as required

A9: CHARGE AIR SHUT-OFF VALVE VACUUM SUPPLY CHECK

	1 Connect the vacuum pump with gauge to the charge air shut-off valve solenoid vacuum supply hose at point TP6
	2 Start the engine and allow to idle
	3 Observe the gauge, a vacuum of approximately 800mbar should be displayed
	Is the correct vacuum achieved and maintained? Yes Remove and inspect the charge air shut-off valve solenoid. Suspect solenoid stuck closed. Replace as required No GO to A18.

A10: TURBINE INTAKE SHUT-OFF VALVE ACTIVATION CHECK 1

	1 Connect a hand operated vacuum pump using a 'T' piece between the turbine intake shut-off valve and the turbine intake shut-off valve solenoid at point TP7
	2 Start the engine and allow to idle. View the vacuum reading while in mono turbo mode (during the first three minutes of idle)
	Is a vacuum present? Yes

Remove and inspect the turbine intake shut-off valve solenoid. Suspect solenoid stuck open. Replace as required

No
GO to A11.

A11: TURBINE INTAKE SHUT-OFF VALVE ACTIVATION CHECK 2

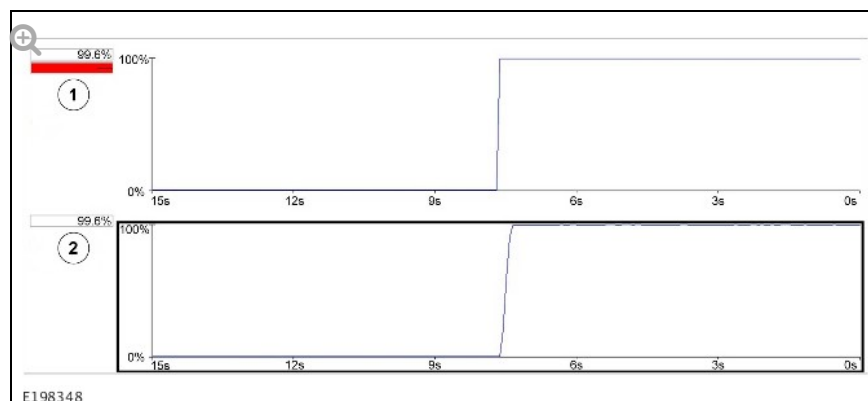
- 1 With the hand operated vacuum pump still connected into the circuit at point TP7
- 2 Allow the engine to idle
- 3 Using the Jaguar Land Rover Approved Diagnostic Equipment, record datalogger signals - **Turbo Shut-off valve opening position - Commanded (0x03F0)** and **Turbo Shut-off valve opening position - Measured (0x03F1)**
- 4 After approximately 3 minutes at idle the turbine intake shut-off valve will automatically be triggered to open and activate the secondary turbo
- 5 This can be confirmed by datalogger signal - **Turbo Shut-off valve opening position - Commanded** changing value to 100%
- 6 When this occurs a vacuum of approximately 800mbar should be displayed on the gauge
- 7 Observe the value on the vacuum gauge as the secondary turbo is activated

Is the correct vacuum is achieved and maintained?

Yes
GO to A12.
No
GO to A17.

A12: TURBINE INTAKE SHUT-OFF VALVE ACTIVATION CHECK 3

- 1 Look at the recorded datalogger signals 1 - **Turbo Shut-off valve opening position - Commanded (0x03F0)** and 2 - **Turbo Shut-off valve opening position - Measured (0x03F1)**



	Is the measured value for the turbo shut-off valve 98% or greater? (See screenshot for reference) Yes GO to A13. No GO to A19.
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A13: TURBINE INTAKE SHUT-OFF VALVE ACTIVATION CHECK 4

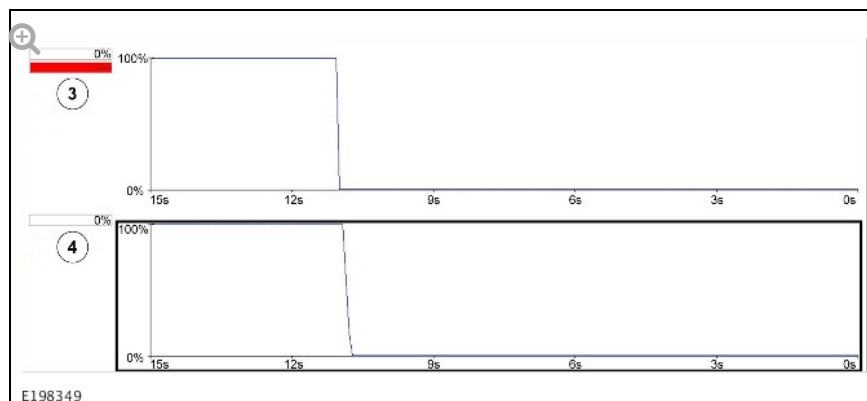
	1 Look at the recording for the datalogger signal 2 - Turbo Shut-off valve opening position - Measured (0x03F1)
	Does the signal show the valve moving smoothly and continuously to its maximum open position? (Refer to screenshot in step A12) Yes GO to A14. No Suspect valve is sticking. Replace faulty turbine intake shut-off valve After installing a new turbine intake shut-off valve, carry out the calibration procedure. Using the Jaguar Land Rover Approved Diagnostic Equipment, perform routine - Powertrain set-up – Turbine shut-off valve

A14: TURBINE INTAKE SHUT-OFF VALVE ACTIVATION CHECK 5

	1 Using the Jaguar Land Rover Approved Diagnostic Equipment, record datalogger signals - Turbo Shut-off valve opening position - Commanded (0x03F0) and Turbo Shut-off valve opening position - Measured (0x03F1)
	2 If the engine speed is increased from idle then the turbine shut-off valve will automatically close returning to mono turbo mode
	3 Increase engine speed and then allow the engine to return to idle
	4 The vacuum will drop back to zero (within 2-3 seconds)
	5 Observe the value on the vacuum gauge as the secondary turbo is deactivated
	Is the correct vacuum achieved and maintained? Yes GO to A15. No Check that the datalogger signal - Turbo Shut-off valve opening position - Commanded (0x03F0) has changed to 0%. If so, remove and inspect the charge air shut-off valve solenoid. Suspect solenoid stuck open. Replace as required

A15: TURBINE INTAKE SHUT-OFF VALVE ACTIVATION CHECK 6

	1 Look at the recorded datalogger signals 3 - Turbo Shut-off valve opening position - Commanded (0x03F0) and 4 - Turbo Shut-off valve opening position - Measured (0x03F1)



Did the signal value for **Turbo Shut-off valve opening position - Measured** change to <4%? (See screenshot for reference)

Yes

GO to A16.

No

Suspect turbine intake shut-off valve is sticking

Install a new turbine intake shut-off valve

After installing a new turbine intake shut-off valve, carry out the calibration procedure. Using the Jaguar Land Rover Approved Diagnostic Equipment, perform routine - Powertrain set-up – Turbine shut-off valve

A16: TURBINE INTAKE SHUT-OFF VALVE ACTIVATION CHECK 7

1 Look at the recorded datalogger signals 3 - **Turbo Shut-off valve opening position - Commanded (0x03F0)** and 4 - **Turbo Shut-off valve opening position - Measured (0x03F1)**

Does the signal show the valve moving smoothly and continuously to its minimum closed position? (Refer to screenshot in step A15)

Yes

No fault present. Exit pin point test

No

Suspect turbine intake shut-off valve is sticking

Install a new turbine intake shut-off valve

After installing a new turbine intake shut-off valve, carry out the calibration procedure. Using the Jaguar Land Rover Approved Diagnostic Equipment, perform routine - Powertrain set-up – Turbine shut-off valve

A17: TURBINE INTAKE SHUT-OFF VALVE VACUUM SUPPLY CHECK

1 Connect the vacuum pump with gauge to the turbine intake shut-off valve solenoid vacuum supply line at point TP8

2 Start the engine and allow to idle

3 Observe the gauge, a vacuum of approximately 800mbar should be displayed

Is the correct vacuum is achieved and maintained?

Yes

Remove and inspect the turbine intake shut-off valve solenoid. Suspect

solenoid stuck closed. Replace as required

No

GO to A18.

A18: VACUUM SUPPLY CHECK

1 Switch off the engine and the ignition

2 With the hand operated vacuum pump still connected into the circuit at point TP8, apply a vacuum

Does it hold vacuum pressure?

Yes

Look for air leaks on the vacuum supply line between the non return valve and the vacuum pump

Carry out smoke test to identify leak point

Check the output pressure directly at the vacuum pump

No

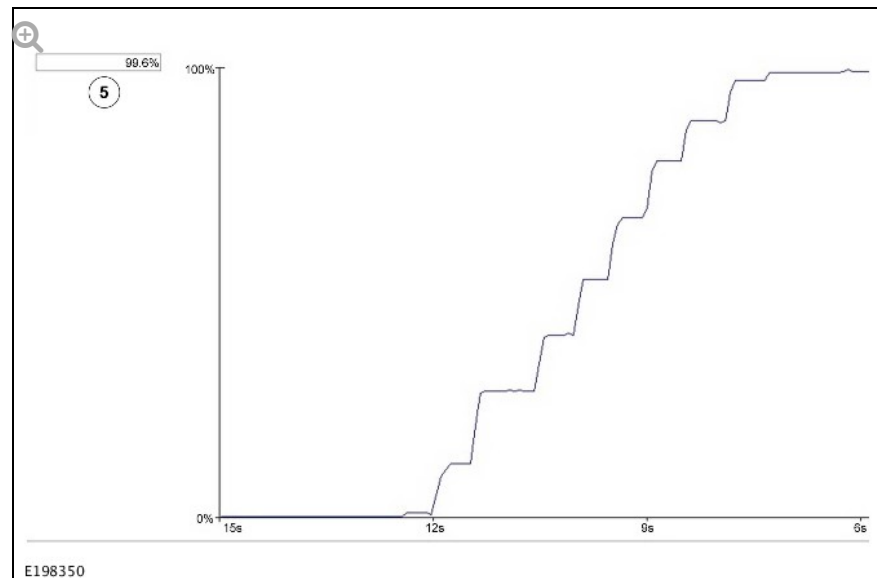
Suspect air leak in vacuum supply line between the non return valve and the turbine intake shut-off valve / charge air shut-off valve solenoids

Carry out smoke test to identify leak point. Repair/replace as required

A19: TURBINE INTAKE SHUT-OFF VALVE OPERATION CHECK

1 Connect a hand operated vacuum pump directly to the secondary turbocharger turbine intake shut-off valve at point TP9

2 Gradually apply a vacuum and observe the turbine intake shut-off valve actuator and monitor the position sensor signal with datalogger



Does the turbine intake shut-off valve operate and is this reflected by signal 5 - **Turbo Shut-off valve opening position - Measured (0x03F1)** value?

Yes

Suspect blockage in vacuum supply line between the turbine intake shut-off valve and charge air shut-off valve solenoid

Inspect hose for defect and rectify as required

No

Turbine intake shut-off valve failure

Install a new turbine intake shut-off valve

After installing a new turbine intake shut-off valve, carry out the calibration procedure. Using the Jaguar Land Rover Approved Diagnostic Equipment, perform routine - Powertrain set-up – Turbine shut-off valve

D T C I N D E X

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: [Diagnostic Trouble Code Index - TDV6 3.0L Diesel - Gen 1.5, DTC: Engine Control Module \(ECM\)](#) (100-00 General Information, Description and Operation).